

Saumya Vijayvargiya

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EDUCATION

- BITS Pilani K K Birla Goa Campus** Goa, India
 - Master of Science - Biological Sciences* *August 2024 - August 2029*
 - Bachelor of Engineering - Electronics and Communication Engineering CGPA: 8.62

PROJECTS

- Aqueous Solubility Prediction** Advised by Prof. Raviprasad Aduri
 - Computational Biology & Drug Delivery | Python, PyTorch, RDKit*
 - Model Architecture:** Developed a Message Passing Neural Network (MPNN) to predict aqueous solubility of drug-like molecules directly from SMILES string representations, leveraging graph-based molecular encoding.
 - Chemical Space Analysis:** Analyzed the chemical space of SMILES-encoded molecules to extract structural and physicochemical insights, enabling a deeper understanding of molecular properties influencing solubility.
 - Data Pipeline:** Engineered a robust data pipeline with an 80/20 train-test split and 5-fold cross-validation on the training set to ensure model generalizability and prevent overfitting.
 - Feature Engineering:** Applied feature importance techniques to identify and select the top chemical descriptors, optimizing model input and improving predictive performance.
 - Benchmarking:** Benchmarked MPNN performance against classical ML models including Random Forest, LightGBM, and XGBoost, demonstrating the effectiveness of graph-based deep learning for molecular property prediction.
- Relevant Courses:**
 - Machine Learning Specialization — Andrew Ng (Coursera)
 - Global Consumer Intelligence (GCI) course by University of Tokyo
 - Bioinformatics | Nanobiotechnology | Genetics | Cell Biology | Microbiology

EXTRACURRICULAR

- Formula Bharat** National Level Competition
 - Core Powertrain Engineer in Team GigaWatt* *2024 - Present*
 - Powertrain System Design:** Designed and integrated mechanical powertrain systems from scratch using CAD tools for a Formula-style race vehicle.
 - Cross-functional Collaboration:** Worked within a high-performance, multidisciplinary team managing tight deadlines and complex system dependencies under competitive pressure.
 - Systems Integration:** Coordinated powertrain components with chassis and electrical subsystems, ensuring reliability and performance under race conditions.
- Department of Creative Works** College Fest Organization
 - Core Member*
 - Actively involved in planning, coordinating, and executing events during college fests as part of the creative works department.

SKILLS

- Languages** Python, C, Java
- ML** Graph-based Neural Networks, Chemical Space Exploration, Regression Modeling, Cheminformatics
- Frameworks** PyTorch, TensorFlow, Scikit-learn, RDKit
- Tools** SOLIDWORKS, Fusion 360, Jupyter Notebook, Google Colab